

Discrete Mathematics and graph Theory

Chapter one: Counting Principles

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Out lines

Topics

In this chapter we will discuss the following topics:

- 1.1. Basic Counting Principles
- 1.2. Permutations and Combinations
- 1.3. The inclusion-exclusion principles.
- 1.4. The pigeonhole principles
- 1.5. The binomial theorem.

Basic Counting Principles

Combinatorics is an important part of discrete mathematics, and is the study of arrangements of objects. Combinatorial analysis is concerned with the methods and principles of determining the logical possibilities of some events without listing and identifying every logical cases. Counting problems arise throughout mathematics, statistics and computer science.

Counting principles enable us to determine

- ☞ the number of ways to perform different tasks,
- ☞ the number of samples space of an experiment,
- ☞ the number possible outcomes of an event.

There are two fundamental counting principles; namely

- ① Addition Principle(AP)
- ② Multiplication Principle(MP).

Addition Principle

Suppose that a task can be performed in n ways, and the second task (unrelated to the first one) can be performed in m ways. Then, the combined task of doing something either from the first or from the second task can be performed in $n + m$ ways.

Example

A university consists only of two faculties, namely Science and Engineering. On Tuesday at 7pm there are 10 exams that can take place from the faculty of Science. At the same time there are 5 exams that can take place from the faculty of Engineering. Then, for this University there are $10 + 5 = 15$ exams that can take place (on Tuesday, 7pm).

Generalization of AP

Suppose that the tasks $T_1, T_2, \dots, T_m$ can be done in $n_1, n_2, \dots, n_m$ ways respectively and no two tasks can be done at the same time. The number of ways to do one of these tasks is

$$n_1 + n_2 + \dots + n_m.$$

Example

A student can choose a computer project from one of the three lists. The three lists contain 23, 15 and 19 possible titles respectively. How many possible projects are there to choose?

Multiplication Principle

Suppose that a task can be performed in n ways, and the second task (unrelated to the first one) can be performed in m ways. Then, the combined task of performing first the first task and then the second task can be performed in $n \times m$ ways.

Example

On a university there are 15 exams that can be scheduled on Tuesday, and 5 (different from the first) exams that can be scheduled on Wednesday. Then on these days there are $15 \times 5 = 75$ different schedules of exams.

Suppose that a first event can occur in n_1 different ways, a second event can occur in n_2 different ways, and so on, and a $m - th$ event can occur in n_m different ways. Then the number of different ways for these m events to occur in succession is given by

$$n_1 \times n_2 \times \cdots \times n_m$$

Example

- 1 If a home security code consists of a sequence of three distinct letters (A to Z), then how many different codes are possible?
- 2 Consider motor vehicle licence plates made up of 3 letters followed by 3 digits, such as ABC012. Determine the total number of possible different licence plates, note that there are 26 choices for each letter and 10 choices for each digit.
- 3 How many numerals each with three digits can be formed using the numbers 1,2,3,4, and 5 so that no digits are repeated?

Exercise

- ① *An office building contains 27 floors and has 37 offices on each floor. How many offices are in the building?*
- ② *There are 18 mathematics and 24 computer science major students at a college. How many ways are there to choose*
 - (a) *one representative who is either a mathematics major or a computer science major?*
 - (b) *two representatives, so that one is a mathematics major and the other is a computer science major?*
- ③ *A room in a building has four doors that are designated as door A, B, C and D. If a person is interested entering to the room by any of these doors and leaving by a different door, then how many possible ways can fulfill his interest?*
- ④ *There are 6 different airlines that fly from Addis Ababa to Washington DC and 7 that fly from Washington to Denver. How many different possibilities are there for a trip from*

Permutations and Combinations

Many counting problems can be solved by finding the number of ways to arrange a specified number of distinct elements of a set of a particular size, where the order of the elements matters and does not matter.

An arrangement of a set of n objects in a given order (where the order of selection matters) is called permutation of the objects (taken all at a time).

Definition (r -permutation)

An arrangement of any $r \leq n$ objects in a given order is called an r -permutation or a permutation of n objects taken r at a time.

Note

The number of permutation of n objects taken r at a time is denoted by $P(n, r)$ or nPr or $P \left(\begin{matrix} n \\ r \end{matrix} \right)$, where $0 \leq r \leq n$.

Example

Consider the set of letters A , B , C and D . Determine the permutation of

- (a) 4 letters taken all at a time.
- (b) 4 letters taken 3 at a time (3-permutation).
- (a) 4 letters taken 2 at a time (2-permutation).

Theorem (Permutations without repeating objects)

The number of permutations of n different objects taken r at a time is given by

$$P(n, r) = \frac{n!}{(n-r)!} = n \times (n-1) \times (n-2) \times \cdots \times (n-r+1)$$

Example

- 1 In how many ways can we select three students from a group of five students to stand in line for a picture?
- 2 In how many ways can three prizes be given for 100 athletes in a race?
- 3 Find the number of permutations of 6 objects taken all at a time.

Note

There are $n!$ permutations of n objects (taken all at a time).

Theorem (Permutations with repeating objects)

The number of r -permutations of n different objects with repetition is $n \times n \times n \cdots \times n = n^r$

Example

- ① How many two digit positive numbers can be formed using 1, 2 and 3?
- ② A multiple-choice test has 100 questions with four possible answers(A, B, C or D) for each question. How many different sets of 100 answers are possible?
- ③ In how many ways can three awards be given to 50 artists, when each artist is eligible for all awards?

Distinguishable Permutations

In this case, we consider permutations of objects some of which are alike.

Theorem (Permutations with alike objects)

The number of permutations of n objects of which n_1 objects are alike and one kind, n_2 objects are alike and another kind, and so on, and n_r objects are alike and another kind is given by

$$P(n : n_1, n_2, \dots, n_r) = \frac{n!}{n_1! n_2! \dots n_r!}$$

Example

- ① How many signals can be formed using 10 flags of which 2 are Red, 3 are Blue and 5 are Yellow?
- ② How many distinct words of length 6 could be formed using the letters of the word ADIDAS?

Circular Permutation

If three persons invited for dinner they may sit in a row in 6 different ways.

The answer would be different if they were to be seated around a circular table, which is exactly in 2 different ways.

Proposition

The number of permutations of n objects around a circle, taken altogether, is given by $(n - 1)!$.

Example

- 1 In how many ways can 4 children sit in a round table?
- 2 In how many ways can 8 boys and 8 girls be seated at a round table for conference so that no two ladies sit together?

Exercise

- ① *How many three positive numerals less than 1000 can be formed from the digits 0, 1, 2, 3 and 4 when repeated digits are possible?*
- ② *A password consists of 4 digits of 1, 2, 3, 4 and 5. What is the largest possible number of unsuccessful attempts in opening the lock is required?*
- ③ *Find the number of distinguishable permutations of the letters in the word **MISSISSIPPI**.*
- ④ *In how many ways can 10 cars be placed in a stock car race if 3 are Chevrolet, 4 are BYD, 2 are Toyota and 1 is Corolla?*
- ⑤ *In how many ways can 8 boys and 7 girls be seated in a dinner around circular table for conference so that no two ladies sit together?*

Combination involves selecting items without considering order (order is of no significant).

Definition

Let $n, r \in \mathbb{N}$ such that $0 \leq r \leq n$. An unordered selection of r elements from an n element set is called a combination.

Example

The following are permutations and combinations of 4 letters a , b , c and d taken three at a time.

Combination	Permutations
abc	abc, acb, bac, bca, cab, cba
abd	abd, adb, bad, bda, dab, dba
acd	acd, adc, cad, cda, dac, dca
bcd	bcd, bdc, cbd, cdb, dbc, dc b

Note

The number of combinations of n objects taken r at a time is denoted by $C(n, r)$ or nCr or $\binom{n}{r}$.

Proposition (number of r -combinations)

The number of combinations of n different objects taken r at a time (r -combinations of n objects) is given by

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \text{ where } 0 \leq r \leq n.$$

Example

1. In how many ways can a committee of five be chosen from a group of 10 members of an association?

Example

2. In how many ways can 11 players be chosen from a group of 13 players if
 - (a) the players are selected at random?
 - (b) a particular player must be included?
 - (c) a particular player must be excluded?
3. A committee of 2 Republicans and 2 Democrats is to be selected from 7 Republicans and 9 Democrats. In how many ways can the committee be formed?
4. How many pairs of dance partners can be selected from a group of 12 women and 20 men?

Corollary

$$C(n, r) = C(n, n - r).$$

Note

A finite set of n elements each with r elements has $\binom{n}{r}$ subsets.

Example

What will be the number of subsets with 3 elements is obtained from a set having 4 elements?

Application of combination: The binomial expansion

Proposition

Let a and b be variables and n be a positive integer, where $0 \leq r \leq n$, then

$$\begin{aligned}(a+b)^n &= \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{n} b^n \\ &= \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r\end{aligned}$$

Example

- ① Find the expansion of

(a) $(a+b)^3$ (b) $(a-b)^4$ (c) $(x+2y)^7$ (b) $(1+2x)^6$

- ② Determine the coefficient of $x^{12}y^{13}$ in the expansion of

(a) $(x+y)^{25}$ (b) $(2x-3y)^{25}$

Note

The $(r + 1)^{st}$ term in the binomial expansion of $(a + b)^n$ is given by

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

Example

- 1 Find the 10th term of $(2x - y)^{11}$.
- 2 Find the middle term of $\left(x - \frac{1}{x}\right)^{10}$.

Exercise

Prove that (a) $\sum_{k=0}^n \binom{n}{k} = 2^n$

(b) $\sum_{k=0}^n \binom{n}{k} 2^k = 3^n$

Pascal's Triangle

Pascal's triangle can be used to calculate the binomial coefficients of an expansion $(a + b)^n$. This is displayed in triangle form as given below.

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & 1 & & 1 & \\ & & 1 & & 2 & & 1 \\ & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \\ & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\ & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \\ & 1 & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1 \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \vdots \end{array}$$

Pascal's Identity: $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$

One can notice that $\binom{6}{4} = \binom{5}{3} + \binom{5}{4}$

Distribution of objects

Proposition

The number of ways of distributing m distinguishable/distinct balls in to n distinguishable boxes(cells), when empty box is allowed is n^m .

Example

Suppose that we have two distinguishable balls a and b and three distinguishable cells(boxes) say 1,2,3. Then the total number of possible distributions of balls to the boxes is $3^2 = 9$. The table of distribution is given as follows.

	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9
Box 1	ab			a	a		b	b	
Box 2		ab		b		a	a		b
Box 3			ab		b	b		a	a

Exercise

Determine the number of distributions of

- (a)** *3 distinguishable balls in to 2 distinguishable boxes.*
- (b)** *4 distinguishable balls in to 2 distinguishable boxes.*

Indistinguishable balls, distinguishable boxes

Proposition

The number of different ways of distributing the n indistinguishable balls into the m distinguishable boxes without exclusion (a box may contain more than one ball), with a possible empty box is

$$\binom{m+n-1}{n}$$

Example

Suppose that there are two indistinguishable balls, and 3 distinguishable boxes. We have the following table of distributions. The number of distribution is 6.

	D_1	D_2	D_3	D_4	D_5	D_6
Box 1	aa			a	a	
Box 2		aa		a		a
Box 3			aa		a	a

Proposition

The number of different ways of distributing the n indistinguishable balls into the m distinguishable boxes without exclusion (a box may contain more than one ball), in such a way that no box is empty is

$$\binom{n-1}{m-1}$$

Example

How many ways are there to put 7 indistinguishable balls into 5 distinguishable boxes?

- (i) If no box is empty
- (ii) If there are empty boxes.

Read about the distribution of Distinguishable balls in indistinguishable boxes.

Derangement

Definition

A permutation of distinct objects in which none of the objects is in its natural or original place is called derangement.

Example

312 is derangement of the numbers 1, 2, 3. But 213 is not.
(Why?)

Notation: The number of derangement of n objects taken altogether is denoted by either the symbol $D(n)$ or d_n .

Example

- 1 List down the derangement of the numbers 1, 2, and 3..
- 2 Find $D(1)$, $D(2)$, $D(3)$ and $D(4)$.

Proposition

The number of derangements of n objects taken altogether at a time is given by

$$n! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \cdots + (-1)^n \frac{1}{n!} \right)$$

Example

$$D(4) = 4! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = .$$

The inclusion-exclusion principle

This principle is a counting technique applied when two or more sets have overlap(or tasks can be done at the same time), since we do not use the sum rule. It uses the method of obtaining the number of elements in the union of sets.

Proposition

Let A , B and C be finite sets. Then

$$(a) \quad n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$(b) \quad n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Example

Determine the number of arrangements of the digits 1, 2, 3, 4, 5, 6 exist, under the constraint that the first digit is 1 or the last digit is 5?

Proposition

Let A , B and C be finite sets, and U is a universal set. Then

$$(a) \quad n((A \cup B)^c) = n(U) - n(A \cup B)$$

$$(b) \quad n((A \cup B \cup C)^c) = n(U) - n(A \cup B \cup C)$$

Example

- ① Suppose that in a group of 100 students; 50 taking mathematics, 40 taking computer science, 35 taking information science, 12 taking maths and computer science, 10 taking maths and information, 11 taking computer science and information science and 5 taking all.
 - (a) How many students are taking at least one subject?
 - (b) How many of them taking none of the subjects?
- ② Determine the number of integers n such that $1 \leq n \leq 100$ that are neither divisible by 3 nor by 5.

The pigeon hole principle

Proposition

If $n + 1$ pigeons are placed into n pigeonholes, then atleast one pigeonhole is occupied by more than one pigeons.

Example

- 1 If a computer department consists of 13 doctors, then atleast two of the doctors were born in the same month.
- 2 In any group of 367 or more people, at least two of them must have been born on the same date.

Theorem

If $kn + 1$ pigeons are placed in to n pigeonholes then at least one pigeon hole is occupied by $k + 1$ pigeons.

Example

Find the minimum number of students in a class to be sure that three of them are born in the same month.

Exercise

- ① *The nominations for officers of the mathematics association of Ethiopia consist of five candidates and three are to be elected. In how many ways can a president, Secretary, and treasurer be elected?*
- ② *There are six points in a plane, no three of which are collinear. In how many ways can you draw a pair of triangles with the six points as vertices?*
- ③ *How many ways can an eight-person committee be chosen from a group of 10 new members and 15 old members if the committee is composed of:*
 - (a) *Four members from each group*
 - (b) *More new members than old members*
 - (c) *At least two new members*
- ④ *How many people come from 5 countries to guarantee that 3 of them come from the same country?*

Thank You!